Shervin Naseri

Senior Machine Learning Engineer | Backend Developer

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PROFESSIONAL SUMMARY

Machine Learning Engineer, Backend Developer and Technical Lead with a strong background in Computer Vision and NLP. Experienced in building backend services, web applications and reliable AI pipelines, including Retrieval-Augmented Generation (RAG), event-driven architectures and LLM tool-calling integrations across Python, Node.js and React. Works across the full cycle, from researching novel architectures to deploying document-processing and forecasting systems in production.

SKILLS

Backend: Python, Node.js, FastAPI, Express, Microservices, APIs, event-driven systems

Frontend: React, Angular, TypeScript, JavaScript, Next.js, Tailwind CSS

AI / LLM & Retrieval: PyTorch, Hugging Face, PEFT, LangChain, OpenAI and Anthropic APIs, AWS Bedrock, Vector databases, Chroma, Weaviate, RAG, prompt design, tool calling, embeddings

Data & Architecture: SQL, NoSQL, PostgreSQL, MongoDB, Redis, Prisma ORM, SQLAlchemy

DevOps & Testing: Docker, CI/CD pipelines, GitHub Actions, Pytest, Vitest, observability, Nginx

Certifications: AWS Certified Generative AI Developer - Professional (Amazon Web Services, 2026-03)

EXPERIENCE

AI & Software Engineer

Independent | Ottawa, ON *2025-03 - Present*

- Architected RAG pipelines and LLM integrations combining vector databases with keyword search for grounded semantic retrieval.
- Built and refactored Node.js and Python services to support structured LLM outputs, schema validation, and tool-calling decoders.
- Engineered asynchronous data processing workflows and optimized PyTorch pipelines for model fine-tuning, embedding generation, and inference.

Graduate Research & Teaching Assistant (part-time)

University of Ottawa | Ottawa, ON *2022-01 - 2024-12*

- Researched the reliability of simulation-based testing for autonomous driving systems; co-authored "Evaluating the Impact of Flaky Simulators on Testing Autonomous Driving Systems" (Empirical Software Engineering; presented at ICSE 2025).

Senior ML Engineer

Pardazesh Etelaat Kian (IraPardaz) | Tehran, Iran *2020-06 - 2021-04*

- Productionized a high-accuracy Farsi document OCR engine handling complex layouts, paragraph blocks, and poetry indentation.
- Built a perceptual similarity search system using embeddings to index, group, and deduplicate real-estate listing images.
- Engineered real-time vehicle detection and tracking using YOLOv4 with perspective-aware regression for traffic analytics.
- Led technical interviews and mentored junior engineers on best practices in model deployment.

Senior Backend Engineer / Tech Lead

Nikaat Corp | Tehran, Iran *2016-12 - 2020-04*

- Architected and maintained RESTful services and backend APIs for Mint and Phosphor portfolios using Node.js, Express, and MongoDB.
- Designed and deployed Bi-LSTM Seq2Seq deep learning architectures for domain-specific currency exchange rate forecasting.
- Engineered a GRU-based encoder-decoder model to predict women's health metrics from sparse, multi-modal time-series data.
- Managed CI/CD deployment pipelines and containerized production environments utilizing Docker and Nginx.

Python Developer

Vestaak | Tehran, Iran *2016-06 - 2016-12*

- Developed a content recommendation engine using Python and Scikit-learn on user interaction data mined from the Telegram API.

PROJECTS

CodebaseMCP (Multi-Tenant RAG & Architecture Indexing) | https://github.com/shervinemp/CodebaseMCP

- Architected an event-driven Model Context Protocol (MCP) server in Python, integrating Weaviate vector search and AST parsing for multi-tenant code retrieval.
- Integrated Gemini APIs with prompt engineering and context grounding to generate structured technical documentation.

ParsiHub: Business Directory for the Persian Community in Canada (pre-launch)

- Built a trilingual (English, Persian, French) platform where people find, review and verify Persian-owned businesses and owners claim and manage listings, on Next.js 16, React 19, tRPC with Zod validation, Prisma and PostgreSQL.
- Implemented hybrid search (pgvector and full-text results fused by reciprocal rank) and a RAG assistant with prompt-injection and PII safeguards; embedding and geocoding jobs run on BullMQ workers with Redis.

VoxPipe (Real-Time Voice & Agentic Tool-Calling Pipeline) | https://github.com/shervinemp/voxpipe

- Engineered an event-driven voice and reasoning system orchestrating streaming ONNX models with LLM tool-calling decoders.
- Built a conversational memory layer using SQLite WAL mode and FTS5 indexing for real-time prompt context injection.

TIDA (Parameter-Efficient Fine-Tuning for Adaptive Computation) | https://github.com/shervinemp/TIDA

- Built a PyTorch LoRA fine-tuning and evaluation pipeline that lets pretrained language models spend extra "thought" tokens on harder tokens (TinyLlama; WikiText and LAMBADA).

EDUCATION

MSc Computer Science

University of Ottawa | *2022-01 - 2025-03*

MSc Software Engineering

Sharif University of Technology | *2018-09 - 2021-10*

BSc Computer Science

Amirkabir University of Technology | *2013-09 - 2018-06*